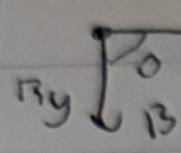
Method of Joints

B



$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{0.3}{0.4}\right)$

$\theta = 36.87^\circ$

$\Sigma F_y = 0 = -By - BC \sin\theta$

$= -8.333 - BC(0.6)$

$BC = -8.333 / 0.6$

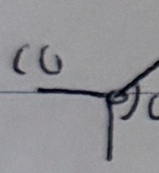
$BC = -13.889\ kN\ (C)$

$\Sigma F_x = 0 = AB + BC$

$= AB - 11.111$

$AB = 11.111 kN$

D



$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{0.3}{0.4}\right) = 36.87^\circ$

$\Sigma F_y = 0 = AD \sin\theta - P$

$= AD(0.6) - 25$

$AD = 25/0.6$

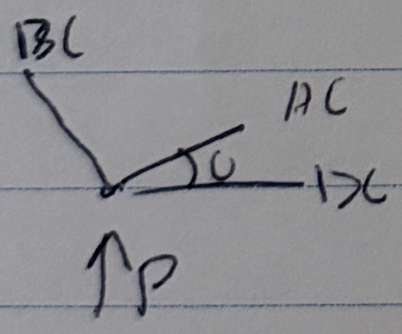
$AD = 41.667 kN$

$\Sigma F_x = 0 = -CD + DA \cos\theta$

$= CD + 41.667(0.8)$

$CD = 33.333 kN$

C



$\theta = \tan^{-1}\left(\frac{0.3}{0.8}\right) = 20.56^\circ$

$\Sigma F_y = 0 = 25 - 8.333 + CA \sin\theta$

$CA(0.3511) = -16.667$

$CA = -47.47 kN$

F_{max} 47.47kN